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To the Editor-in-Chief
Processes (MDPI)

Subject: Submission of a review manuscript entitled "Critical Analysis of Quantum-Inspired Evolutionary Algorithms for Multi-Objective Optimization: Taxonomy, Features, and Limitations"

Dear Editor,

We are pleased to submit the above manuscript for consideration as a Review article in *Processes*.

Quantum-inspired evolutionary algorithms (QIEAs) represent candidate solutions as probabilistic Q-bit chromosomes updated by rotation-gate operators. They are increasingly used to solve engineering optimization problems on classical hardware. Most real process-design and operation problems are multi-objective, but the multi-objective branch of the QIEA literature has never been surveyed on its own. It has been active since at least 2003, but it remains small and fragmented. Our manuscript addresses this gap with four contributions:

1. A bibliometric analysis of 2,250 QIEA papers (1996–2026) built from OpenAlex. Multi-objective work makes up under a tenth of the field, and its share has roughly doubled in recent years.

2. A five-axis taxonomy of multi-objective QIEAs: solution representation, quantum-inspired operator type, selection mechanism, hybridization with classical multi-objective evolutionary algorithms, and application domain. We apply it in a systematic feature comparison, with single-objective QIEAs as a baseline.

3. A critical analysis of the field's limitations. Selection mechanisms vary widely, but none is reference-point- or hypervolume-based. Several widely cited methods do not state how they aggregate their objectives, and systematic benchmarking is scarce.

4. An illustrative case study: a block-partitioned quantum-circuit-optimization pipeline extended with NSGA-III and SMS-EMOA. On a 225-run experimental baseline, it shows how an adaptive hypervolume reference point can change a cross-algorithm ranking, which is a concrete benchmarking pitfall.

We believe the manuscript fits the scope of *Processes*, in particular its "Process Control, Modeling and Optimization" and "Automation Control Systems" (optimization algorithms) sections, and it is relevant to "Energy Systems". The survey is intended as a guide for practitioners who choose or design evolutionary optimizers for process problems. To connect its findings directly to process engineering, Section 7 works through two hypothetical but representative application scenarios: the multi-objective design of a methanol plant and the 96-step dispatch of an industrial microgrid. It shows where the identified limitations matter most: mixed continuous/discrete variables, hard constraints, and expensive evaluations. Finally, the manuscript outlines future research directions within FADSE, an open-source framework for automatic design space exploration with parallel evaluation on high-performance computing systems.

To support reproducibility, the pipeline code, the experiment tooling, the 225-run dataset, and the cached bibliometric data are openly available, as described in the Data Availability Statement.

We confirm that neither the manuscript nor any parts of its content are currently under consideration for publication with or published in another journal. All authors have approved the manuscript and agree with its submission to *Processes*.

This manuscript has not been submitted previously to any MDPI journal.

Thank you for considering our work. We look forward to hearing from you.

Sincerely,

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